Alex Rop

Geospatial Engineer · Founder & Director, ROXR Earth

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PROFILE

Geospatial Engineer and founder of ROXR Earth, using satellite intelligence to support the people and places most exposed to environmental risk: drought risk from region to farm, and where degraded land can be restored. I combine Earth Observation, GIS and machine learning in reproducible workflows that turn imagery into practical information for climate risk and land decisions across Africa. Based in Nairobi, working with international partners remotely and with communities and stakeholders in the field.

EXPERIENCE

Fellow, Empower Africa Programme

Sep 2026 – Present

United Nations Institute for Training and Research (UNITAR) REMOTE

- Fellow of the Empower Africa Programme (TICAD9), UNITAR's Japan-Africa public-private partnership programme, with ROXR Earth as my venture. Phase I completed (entrepreneurship, AI and digital transformation, business model and investment readiness) with a full business blueprint and pitch; selected for Phase II (November 2026).

Founder & Director

Jun 2026 – Present

ROXR Earth Limited · Nairobi · roxr.earth

- **Drought risk (live at county level):** designed and operate the ROXR Climate Intelligence Index (RCII), a reproducible Python and Google Earth Engine pipeline that weighs satellite vegetation, heat and rainfall signals against the people and livelihoods exposed, producing a monthly risk level, recommended action and 30/60/90-day outlook for ten ASAL counties; tested against Kenya's official drought assessments (KFSSG and IPC).
- **Farm-level intelligence (in testing):** extended the same reading to sub-county, ward, 5 km grid and farm level, with pre-registered tests of farm-level indicators on harmonized Landsat and Sentinel-2 imagery.
- **Dynamic land suitability (in testing):** built a parallel product that reads, site by site, where rangeland reseeding can succeed and when to sow, and where water pans will fill, hold water and are most needed, re-read as conditions change.
- **Delivery and company:** monthly briefs, an interactive dashboard and structured data feeds built for insurers, lenders, adaptation finance and humanitarian teams, with automated regression testing and a documented record of every change; lead partnerships, funding applications, data licensing and governance.

Geospatial & Stakeholder Engagement Consultant

Aug 2025 – Present

Futures Inc. (Japan) · Lake Victoria, Kenya HYBRID

- GIS and remote sensing for spatial site selection and land use assessment on the Yamaha Carbon Credit Project, a nature-based carbon project, coordinating across international stakeholders with field visits to meet local stakeholders.

Geospatial Data Analyst Consultant

Jan 2025 – Feb 2026

Fairtrade Africa · Eastern and Central Africa HYBRID

- Led land cover change detection and deforestation risk monitoring across 63+ Producer Organizations in six East and Central African countries, using OpenStreetMap, satellite imagery and thematic maps.
- Managed validation and quality assurance of monitoring data used in EU Deforestation Regulation (EUDR) compliance reporting.
- Developed and delivered geospatial data interpretation training to field teams, virtually and in person, improving data collection quality and spatial literacy among non-technical staff.

Geospatial Research Assistant & SDL Intern

Aug 2024 – May 2025

Harvard University, Center for Geographic Analysis REMOTE

- Developed automated Python and KNIME workflows integrating MODIS, Landsat, CHIRPS and soil moisture data across a 23-year time series for multi-source drought index monitoring, including drought analysis in India. Supervised by Devika Jain, AI and Data Science Manager.

Internship Trainee

Jan 2024 – Mar 2024

Esri Eastern Africa · Nairobi

- Built a disaster risk prototype for Kenya's ASAL counties combining suitability modelling, Random Forest drought forecasting and ArcGIS Dashboard visualization for government and humanitarian stakeholders.

Freelance Geospatial Engineer

Feb 2023 – Nov 2023

Uasin Gishu County · Eldoret

- GIS consultancy including work with the Clerk of Works at Kapseret CDF, supporting spatial data management and mapping for county-level infrastructure and development planning.

Attaché

Jan 2022 – Apr 2022

Ministry of Lands and Physical Planning · Eldoret

- Led data collation for the Nine Farms Survey Project, a multi-agency cadastral survey with the Uasin Gishu County Government and the national Ministry of Lands; took part in land subdivision surveys, land dispute resolution fieldwork, topographic surveying and boundary verification.

PUBLICATIONS & PRESENTATIONS

Jain, D. & Rop, A. (forthcoming). *Harmonizing Earth Observation Data for Enhanced Spatial Analysis.* Book chapter, Elsevier. A KNIME-based workflow for multi-source Earth Observation data integration, with a case study on surface urban heat islands in Phoenix, Arizona.

Automated CHIRPS Rainfall Data Extraction for Multi-Source Drought Index Monitoring. Team presentation, Harvard CGA Conference, 2025.

EDUCATION

B.Eng, Geospatial Engineering · Second Class Honours (Upper Division)

2016 – 2021

The Technical University of Kenya

- Final-year project: mapping changes in built-up areas, with Mombasa Road, Nairobi as the case study.

TECHNICAL SKILLS

Programming	Python (pandas, NumPy, SciPy, scikit-learn, GeoPandas, Rasterio, GDAL, matplotlib, Folium), R, JavaScript for Google Earth Engine, KNIME
Earth Observation	Google Earth Engine, MODIS, Landsat, Sentinel-2, CHIRPS; anomaly detection, time-series analysis, land cover classification
Machine learning	Random Forest classification, time-series cross-validation, feature engineering, class imbalance handling, feature importance analysis
GIS & spatial data	QGIS, ArcGIS Pro, PostGIS/PostgreSQL, coordinate reference systems, zonal statistics, spatial data management
Data formats	GeoTIFF, NetCDF, Shapefile, GeoJSON, CSV
Reproducibility	Git/GitHub, pipeline run logging, documented methodology, reproducible workflow design

TRAINING & CERTIFICATIONS

- Spatial Data Science: The New Frontier in Analytics · Esri, 2024
- GIS for Climate Action · Esri, 2024
- AiCE, Career Essentials · ALX, 2024
- Machine Learning for Everyone · DataCamp, 2021
- AI Fundamentals: Core Concepts and Principles · edX
- AI in Society: Ethics, Leadership, and the Road Ahead · edX
- From Space to Screen

LANGUAGES

English (full professional) · Kiswahili (native) · Kalenjin (professional working)

REFERENCES

Available on request.